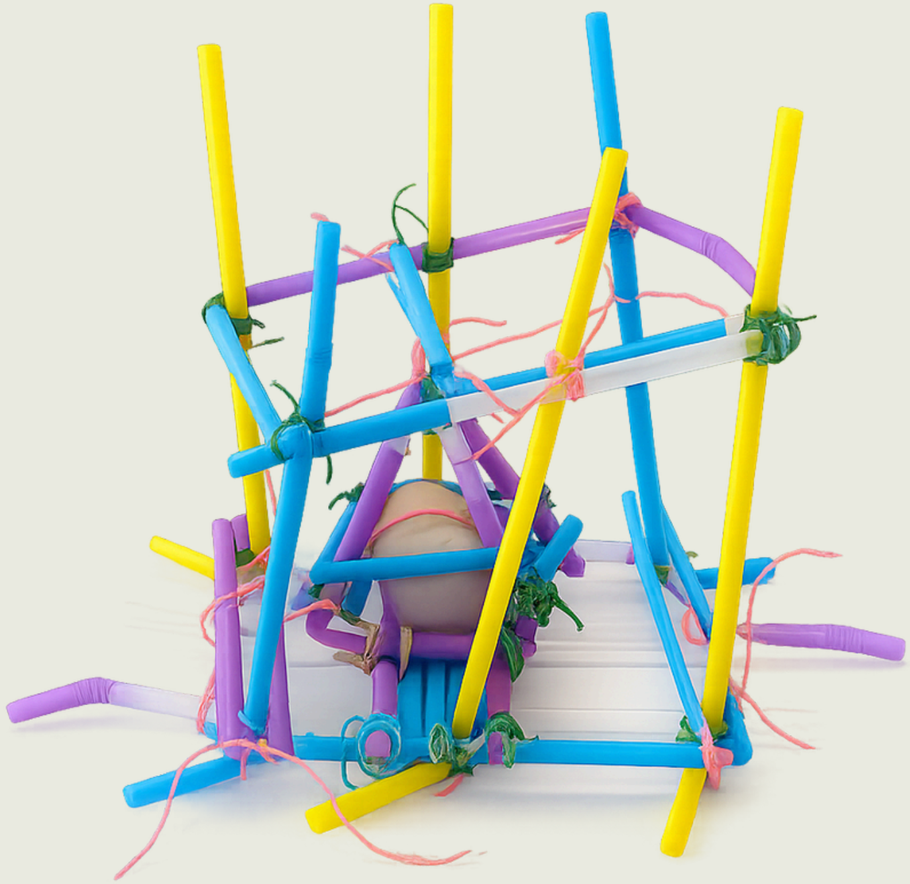
DESIGN MILESTONS

EGG DROP CHALLENGE

FIRST EGG DROP CHALLENGE

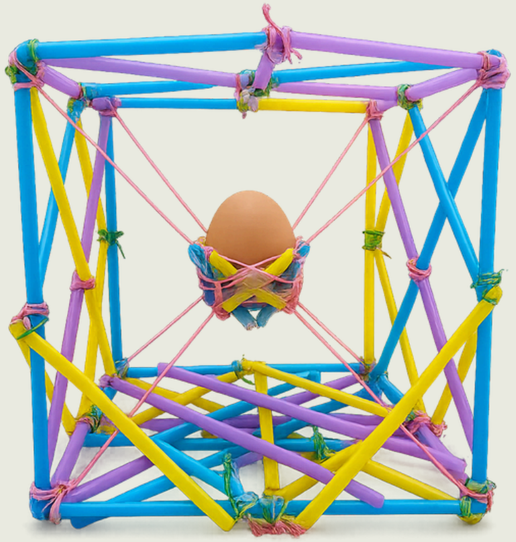


I built this model without any AI assistance, and as a result, it didn't break from 1.5 meters, but it cracked from 3 meters.

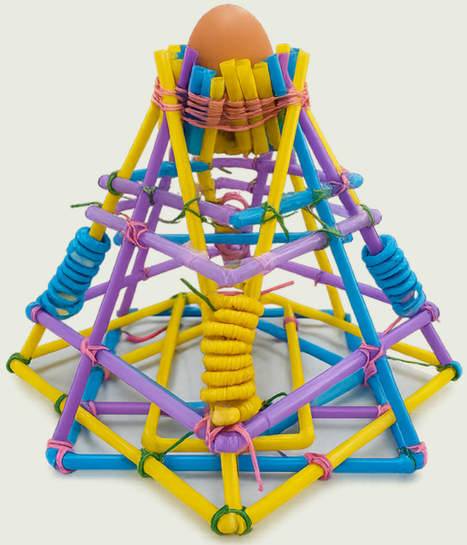
PROMPT

In the last lesson, I created the model in the picture for the egg drop challenge using straws and string. It **didn't break from 1.5 meters**, but it **cracked a little from the 1st floor**, that is, from 3 meters. Because the egg's weight was heavy for protection beneath, it hit the ground and cracked. Now I need to create a new model, but this time there are **material limitations**: I can use **20 straws, 1 meter of tape, and 2 meters of string**. The maximum size also needs to fit within a **15x15x15** cube. **What kind of model should I make to prevent the egg from breaking?**

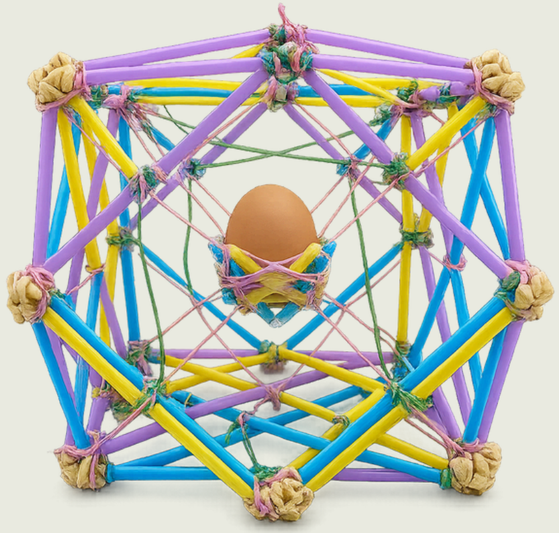
AI RESPONSE



In this design, the egg will never touch the ground or external straws, thus solving the "weight being too much for protection" problem found in the previous model.



In this design, we place the egg on top and create a "crumple zone" underneath it using straws that sequentially crush the egg upon impact.



In this model, we'll secure the egg precisely in the center, but instead of hanging it with strings, we'll create rigid but crushable straw cages both above and below it. This way, no matter which direction the model falls, a "crumple zone" will absorb the impact.

FINAL EGG DROP CHALLENGE

Before starting this model, I consulted AI, but since I didn't find their ideas reliable, I wanted to build upon the model I made for the first egg drop challenge. I thought it would be logical to add a little more string and straws around the nest part of the first model. As a result, my model was successful.

MY REFLECTION

- I completed the **first model without any AI assistance**. In the second model, I was asked to use AI as a partner, and I used AI in the roles of **Re-framer** and **Critique**. I wrote a clear and concise prompt, but the ideas it offered didn't seem logical or reliable to me. Therefore, I decided to proceed with my own idea based on my previous model, and **the result was successful**. I think AI is a helpful tool, but it didn't contribute much to this project.